

الإجابات على "كراس الإجابة"

Treat the following questions:

I- Choose the correct answer: (1pt/correct answer)

1- Class diagram :

- a- Is not the most important in object-oriented development.
- b- Is a static diagram.
- c- Represents the physical model of features.
- d- None of the above.

2- The chronological succession of operations carried out by an actor is represented by:

- a- Class diagram
- b- Use case diagram
- c- Sequence diagram
- d- Another kind of diagram

3- What is a so-called extend relation?

- a- Use case X may eventually need use case Y.
- b- Use case X is done before use case Y.
- c- Use case X always needs use case Y.
- d- Use case Y may eventually need use case X.

4- Between a Vehicle class and a Wheel class, what type of relation is adequate?

- a- Composition
- b- Association
- c- Inheritance
- d- Aggregation

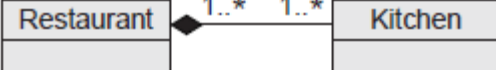
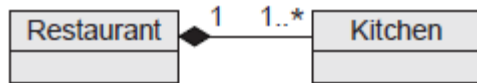
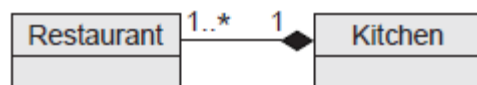
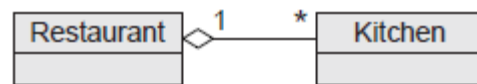
5- Consider an association between a Customer class and an Order class. What multiplicity would you put on the Client's side?

- a- 0..1
- b- 0..*
- c- 1..*
- d- 1..1

6- Which of the following statements about the combined fragments in the Sequence diagrams is true?

- a- A combined fragment is a rectangle with a small pentagon in the upper left corner that should contain the construct keyword.
- b- A combined fragment necessarily consists of several sub-fragments.
- c- Some combined fragments can control the flow of action.
- d- None of the above

7- How do you model the following situation with a class diagram: Every restaurant has at least one kitchen, one kitchen is part of exactly one restaurant?

- a- 
- b- 
- c- 
- d- 

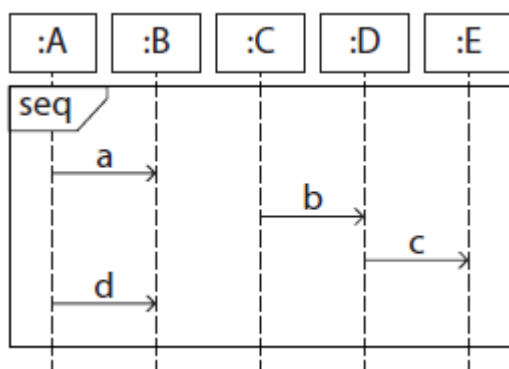
8- Which of the following statements about compositions (composite aggregations) are false?

- a- When the composite element is deleted, the parts also die.
- b- In a composition, a part may belong to only one composite at a time.
- c- The multiplicity of a composite aggregation may be ≥ 1 .
- d- The composite aggregation is a transitive relationship.

9- The association between an actor and a use case:

- a- Is modeled with a dashed line.
- b- Might have multiplicities.
- c- Might be ternary.
- d- Is modeled with a directed edge with the arrowhead pointing to the use case.

10- In the following sequence diagram, which traces are possible?



- a- $a \rightarrow d \rightarrow b \rightarrow c$
- b- $a \rightarrow d \rightarrow c \rightarrow b$
- c- $a \rightarrow b \rightarrow c \rightarrow d$
- d- $d \rightarrow a \rightarrow c \rightarrow b$

II- Open Questions : (1pt/ question)

1- Give the sequence diagram related to description: " Money Withdrawal" which has the following elements:

- a- The teller enters the customer's account number;
- b- The application validates the account with the central system;
- c- The teller requests a withdrawal of 200,000LBP;
- d- The "counter" system queries the central system to ensure that the account is sufficiently funded;
- e- The central system debits the account;
- f- In return, the system notifies the teller that he can issue the requested amount.

2- Draw the use case diagram:

In a school establishment, we want to manage the reservation of classrooms as well as educational material (laptop or / and video projector).

- a- Only teachers are authorized to make reservations (room or equipment availability).
- b- The planning of the rooms can be consulted by everyone (teachers and students).
- c- On the other hand, the hourly summary per teacher (calculated from planning of rooms) can only be consulted by the teachers.
- d- Finally, for each training, there is a responsible teacher who alone can edit the timetable for the entire training.

3- Draw the use case diagram to meet the following expected results:

The person in charge of the football championship wishes to have:

- a- A list of players per team in a given season in order to verify the legal situations.
- b- A list of coaches and co-coaches per team for a given season.
- c- A complete history of all the matches made in a given season.

4- What are the various components in sequence diagrams?

5- What are subclass and supper class?

6- Make the class diagrams corresponding to the following 2 cases:

- a- A directory contains files.
- b- A stock market transaction is a buy or sell.

7- Explain 4 types of diagrams in UML.

8- Consider the following system:

A flight connects two airports (origin and destination). An airport is used by many flights. Develop the corresponding class diagram. Identify classes and relations, indicate any constraints and indicate the multiplicities of each relation.

9- What is the generalization and the specialization?

10- Sequence diagram:

When an e-mail is sent by the sender, the sender does not want to wait for the recipient to receive it and there is no intermediary. Can we use a synchronous message? Complete the figure below with arrows representing messages.



Is transmitting an operation or a signal? In all cases, give elements of a class diagram consistent with the sequence diagram.